
board P
4S3P 18650 cells, about 145 Wh (0-60)
F1 25 A blade
F2 SCF9510 chemical fuse
Q1 charge, Q2 discharge FETs, BQ40Z50 primary, BQ77207EE secondary
MAIN_P
XT60, 12 AWG
board E
vehicle or shore 9 to 36 V
F3 10 A, LM74700, LMS0069 hot swap, 6.15 A declared
VIN_RAW
solar
F2 10 A, LTC7654 tracker (bench-fitted), 6.16 A, OReil
E: F3 25 A to CELL_F
CELL_F
APE3205 5 V, always on: sensor RP2040, sensors, fan logic
dock pins 1 to 4, 12.31 A declared
dock block, 4 x 9 A pins
board A
U2 LMS176 front end, 20 V, U16 restart guard
U3 BQ25731 charger, 4S, no BATFET
CELL+
F1 25 A
R17 5 mOhm charge shunt
VBAT = VSYS 10.0 to 16.8 V
U12 TPS62933: +3V3 logic
U15 LMS176: +12V_HF
U4, U6 APE4500: +5V_S1, +5V_S3
B: slots 1 and 3
U5 LMS176: +5V_S2, 7.2 A minimum limit
B: slot 2
U21 eFuse: monitor
U23 eFuse: +5V_DB
U22 eFuse: Glenair port VBUS
B: hubs, supervisors, radios, KS2, PANEL_5V to C
U7 LMS176: +5V_DEV, 7.2 A minimum limit
U16 LMS176 boost: +54V_POE
B: PoE out
U19 LMS176 and U18 TPS2529 RDA: USB-C PD out, power only
U22 eFuse, U33 12.0 V buck: heater mat
U13 LMS176: +13V8_PA, its own average current loop